



Medius Bridge

Simple setup guide for VisionLabs

FIRMWARE 3.4.1 • PROTOCOL 8

Start with the hardware in your hand. Finish with a working setup.

Before you start

You need a MAKCU/Medius box, a Windows PC with VisionLabs installed, and the latest Medius Bridge plugin package. If you are using a console through a XIM, use the wiring diagram on the next page.

Important: Medius firmware 3.4.1 uses protocol 8 and library ABI 8. Update the firmware first, then install this plugin so VisionLabs installs the matching library.

1. Update the firmware

Open the official Medius dashboard: medius.k4tech.net/dashboard/update. Connect the box to the PC, select the update, and wait until it completes. Do not unplug the box during the update.

2. Connect the control cable

For a normal PC setup, connect MAKCU USB 2 (the control/serial port) to the PC running VisionLabs. Windows should detect a CH343 COM port.

3. Install the plugin

In VisionLabs, open Plugins → Install and select **Medius_Bridge.vlplugin**. Enable Medius Bridge and approve the permission prompt.

4. Let VisionLabs finish setup

The first start may take longer while VisionLabs creates the plugin environment and installs the Medius library. Wait for the plugin log to show **medius lib 3.4.1, abi 8** and **connected fw 3.4.1 proto 8**.

Console setup through XIM

Use this exact physical layout. The port numbers below are the ports on each device.

MAKCU / MEDIUS	XIM
USB 1 → XIM	Port 1 ← MAKCU / MEDIUS
USB 2 → PC running VisionLabs	Port 2 ← Keyboard
USB 3 ← Mouse	Port 3 ← Xbox controller
	Port 4 ← Xbox / console

Signal path

Mouse	MAKCU USB 3	MAKCU USB 1	XIM Port 1	Console
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The keyboard and controller connect directly to the XIM. USB 2 is only the control link to the VisionLabs PC; it is not the game signal path.

First launch checklist

Complete these checks in order. Most setup problems are caused by one cable in the wrong port or an old cached Python environment.

1. Confirm the box is powered and the control cable is connected to the VisionLabs PC.
2. Start VisionLabs and enable Medius Bridge in the Plugins page.
3. Open the plugin settings. Leave Serial port blank for automatic detection unless you have more than one box.
4. Choose an activation mode. The default uses the physical right mouse button to activate while held.
5. Start with Output source = This plugin. The plugin reads VisionLabs tracking and sends relative mouse movement through Medius.
6. Test with the overlay enabled. A green status means the plugin is active; amber means it is waiting for a valid target or activation.
7. Move the mouse and press the activation button. The game/console should receive the physical mouse input through the box, with plugin movement layered on top.

Healthy connection: the log should contain **connected fw 3.4.1 proto 8**. If it says protocol 7, ABI 7, NO_REPLY, or BAD_PROTO_VER, continue to the troubleshooting page.

What 3.4.1 adds

The updated Medius release adds packet-based CLIP triggers, CLIP support for pan/raw reports/control transfers, better device-surface compatibility, report-rate timing improvements, and stability fixes. The Bridge plugin continues to expose the simple mouse/vision workflow; users do not need to configure the developer-level CLIP features to use the plugin.

Troubleshooting

No device found

Try a different USB port and confirm USB 2 is connected to the VisionLabs PC. If you have more than one box, choose its COM port in the plugin settings.

Protocol mismatch / connection drops

Update the box at the Medius dashboard, then rebuild the VisionLabs plugin environment. Quit VisionLabs, delete the environment folder `<VisionLabs>\PythonEnvslcom_staybeaming_medius_bridge`, and start VisionLabs again.

The plugin still installs an old Medius library

Check the Python setup log. Reinstall the plugin package so VisionLabs re-reads requirements.txt. The installed library must report **medius lib 3.4.1, abi 8**.

No movement in the game

Confirm activation is on, the target meets the confidence/class filters, and the correct output source is selected. Start with Gain 0.25, Deadzone 0, and a conservative Max counts per step.

XIM setup does not work

Power down the console and re-check every port against the diagram. MAKCU USB 1 must go to XIM Port 1; keyboard/controller/console belong on XIM Ports 2/3/4.

Need more help? Include the last 20 lines of the Medius Bridge log, your VisionLabs version, and whether you are using a direct PC setup or XIM console setup. Do not include passwords or private account details.